

# Quantitative Research Methods

## Chapter 7: Moving Beyond Linearity

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# The course at a glance

| Chapter      | Topic                                                                 |
|--------------|-----------------------------------------------------------------------|
| Ch. 1        | Introduction                                                          |
| Ch. 2        | Statistical learning — accuracy, bias–variance, Bayes classifier, KNN |
| Ch. 3        | Linear regression — estimation, inference, diagnostics                |
| Ch. 4        | Classification — logistic regression, confusion matrix, ROC           |
| Ch. 5        | Resampling — validation, CV, bootstrap                                |
| Ch. 6        | Model selection and regularisation — ridge, lasso                     |
| <b>Ch. 7</b> | <b>Moving beyond linearity — splines, GAMs</b>                        |
| Ch. 8        | Tree-based methods — trees, forests, boosting                         |
| Ch. 10       | Deep learning — MLPs, CNNs, training                                  |

Nine chapter decks of 180 minutes each, after a taught precourse session; Chapters 2, 3 and 4 run over two sessions apiece. Today's chapter is highlighted. Short exercises every ~20 min; extended exercises every ~45 min; each chapter has a companion lab notebook.

Support vector machines (Ch. 9), survival analysis (Ch. 11), multiple testing (Ch. 13) and unsupervised learning (Ch. 12) are optional self-study modules in Chapters/Advanced/.

# Contents

- 1 Why this chapter matters
- 2 Polynomials
- 3 Splines
- 4 Smoothing
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Optional and advanced material is collected in the **appendix** at the end of the deck.

# Notation in this chapter

| Symbol              | Meaning                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------|
| $b_j(x)$            | Basis function: fixed transformation of $x$ entering linearly                           |
| $d$                 | Degree of a polynomial (or of each spline piece)                                        |
| $C_k(x)$            | Indicator (dummy) for the $k$ -th bin of a step function                                |
| $\xi_k$             | Knot: breakpoint where adjacent spline pieces join                                      |
| $(x - \xi)_+^3$     | Truncated power term: 0 for $x \leq \xi$ , else $(x - \xi)^3$                           |
| $K$                 | Number of knots; cubic-spline $\text{df} = K + 4$ , natural-spline $\text{df} = K$      |
| $\lambda$           | Smoothing parameter of the penalised (smoothing-spline) fit                             |
| $\int g''(t)^2 dt$  | Roughness penalty: total curvature of the fit $g$                                       |
| $\text{df}_\lambda$ | Effective degrees of freedom, $\text{trace}(S_\lambda)$ , where $\hat{g} = S_\lambda y$ |
| $s$                 | Span: fraction of nearest neighbours used by LOESS                                      |
| $f_j(X_j)$          | Smooth component function of predictor $X_j$ in a GAM                                   |

# Sources and references

## Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer.

## About these slides

Each method is introduced with motivation, intuition, formal definition, and worked examples. Slide titles are descriptive.

# Learning objectives for this chapter

By the end of this chapter you will be able to:

- **Fit** polynomial and step-function regressions and discuss their respective failure modes.
- **Construct** regression splines, smoothing splines, and local regressions; **choose** smoothness by cross-validation.
- **Articulate** the boundary problem of cubic splines and the fix offered by natural splines.
- **Build** a generalised additive model (GAM) over multiple predictors and **interpret** its component functions.
- **Reason** about when nonlinear methods help and when they merely add variance.

# Roadmap of this chapter

## A ladder of flexibility

- 1 Polynomials.
- 2 Step functions.
- 3 Regression splines (cubic, natural).
- 4 Smoothing splines.
- 5 Local regression (LOESS).
- 6 Generalised additive models (GAMs).

All of these are linear regression with cleverly chosen basis functions.

# Outline

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- 2 Polynomials
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# Why move beyond linearity?

Linear regression is interpretable and stable, but the world is rarely linear. Whenever the relationship between  $X$  and  $Y$  bends, a linear model leaves bias on the table.

## The strategy of this chapter

Replace the strict linear form  $\beta_j X_j$  with smooth functions  $f_j(X_j)$ , while keeping the additive structure that makes linear regression so easy to interpret.

# Three reasons nonlinearity matters

## Concrete examples

- Wage data peaks in middle age and then drops — not linear.
- Sales have diminishing returns to advertising spend — not linear.
- Survival probability is bounded in  $(0, 1)$  — not linear.

## Takeaway

Every method in this chapter is just linear regression with a cleverly chosen *basis expansion* of the original predictors. The OLS machinery still applies.

## Live demo — §1 of the notebook

Switch to `chapter_07_lab.ipynb` §1 and run the two polynomial cells: a raw-power degree-4 OLS on Wage ( $R^2 \approx 0.086$ ) whose grid plot shows exactly that middle-age wage hump.

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# Polynomial regression: the simplest extension

Replace  $X$  with a basis of powers.

## Model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon.$$

## How to read this

Still a *linear* model in the coefficients — all OLS machinery applies (standard errors,  $F$ -tests, CV). We are simply augmenting the design matrix.

# Polynomial regression: the boundary problem

## Beware of high degree

Degrees beyond 4 rarely improve fit but routinely produce wild oscillations near the boundaries of the data range. Splines are the safer alternative.

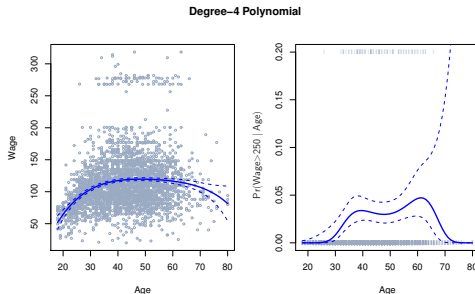
## Why the wiggles?

A single global polynomial must use *one* formula everywhere. Each extra degree buys another turning point, and where data are sparse (the edges) those bends are barely constrained — so the curve swings and its confidence band balloons.

## Takeaway

A useful tool for capturing mild curvature, especially with degree 2 or 3. Beyond that, switch to splines.

# Wage data: degree-4 polynomial



## Takeaway

Left: the degree-4 fit with its pointwise band; right: the fitted  $\Pr(\text{wage} > 250 \mid \text{age})$ . The curve captures the rise-then-plateau, but bends and widens at the sparse upper edge.

## Step functions: piecewise constant

An alternative when the relationship is naturally piecewise constant.

### Model

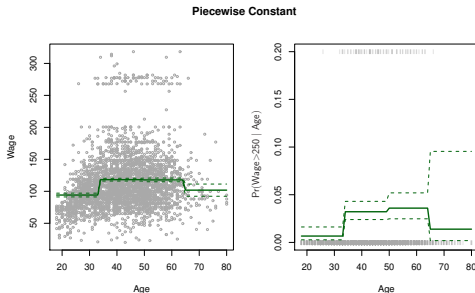
Bin  $X$  into  $K + 1$  ranges and fit a constant in each:

$$Y = \beta_0 + \beta_1 1\{c_1 \leq X < c_2\} + \cdots + \beta_K 1\{c_K \leq X\} + \varepsilon.$$

### Pros and cons

Easy to interpret (“income bracket”), but discontinuous at the breakpoints — often visually ugly. Useful when natural categories exist (age groups, salary tiers), or when banding is chosen deliberately for auditability — at the price of a visible jump for two cases either side of a cut-off.

# Step function fit on Wage



## Takeaway

Piecewise-constant: flat within each band, jumping at every cutpoint. The level shifts are captured, but the discontinuities are an artefact of binning.



## Exercise 7.1 — Evaluating a fitted cubic [Math]

### Exercise

Consider a cubic polynomial regression of wage on age using the raw basis  $\{1, x, x^2, x^3\}$ .

- 1 Write the design-matrix row for a single observation with  $x = 2$ .
- 2 The fitted model is  $\hat{f}(x) = 50 + 4x - 0.5x^2 + 0.1x^3$ . Predict the response at  $x = 2$  and  $x = 4$ .
- 3 Degree three sounds nonlinear. Explain why this is still a *linear* model.

## Solution — Exercise 7.1 (1/2)

## Solution

**(a) Design-matrix row.** Polynomial regression is ordinary OLS on a design matrix whose columns are powers of  $x$ , so each observation contributes the row of its basis values. With basis  $\{1, x, x^2, x^3\}$  and  $x = 2$ :

$$[1, 2, 2^2, 2^3] = [1, 2, 4, 8].$$

For  $n$  observations the full design matrix stacks such rows:  $X_{i\cdot} = [1, x_i, x_i^2, x_i^3]$ .

**(b) Predict by substituting term by term** into  $\hat{f}(x) = 50 + 4x - 0.5x^2 + 0.1x^3$ :

$$\hat{f}(2) = 50 + 4(2) - 0.5(4) + 0.1(8) = 50 + 8 - 2 + 0.8 = 56.8,$$

$$\hat{f}(4) = 50 + 4(4) - 0.5(16) + 0.1(64) = 50 + 16 - 8 + 6.4 = 64.4.$$

*Interpretation:* between  $x = 2$  and  $x = 4$  the prediction rises by  $64.4 - 56.8 = 7.6$ ; the curve still climbs here, but the negative  $x^2$  term is already damping the growth.

## Solution — Exercise 7.1 (2/2)

### Solution

**(c) Why this is still a linear model.** “Linear” refers to the *coefficients*, not to the shape of the fitted curve. Writing

$$\hat{f}(x) = \beta_0 \cdot 1 + \beta_1 \cdot x + \beta_2 \cdot x^2 + \beta_3 \cdot x^3,$$

we see  $\hat{f}$  is a linear combination of *known, pre-computed* regressors  $1, x, x^2, x^3$ . Hence all of the OLS machinery applies unchanged: closed-form  $\hat{\beta}$ , standard errors,  $t$ - and  $F$ -tests, cross-validation. The nonlinearity lives in  $x$ , not in the parameters.

*Interpretation:* this is the basis-function idea behind the whole chapter — keep the linear-regression machinery, change only the columns of the design matrix.

### Common mistake

Predicting  $\hat{f}(4)$  by doubling  $\hat{f}(2)$ . Polynomial predictions do not scale linearly in  $x$ ; every power term must be substituted separately.

## Exercise 7.2 — Step functions and dummy coding [Concept]

### Exercise

We fit a piecewise-constant (step-function) regression of wage on age using cutpoints at 35 and 50, giving three bins:  $(-\infty, 35)$ ,  $[35, 50)$ ,  $[50, \infty)$ .

- 1 How many dummy variables do we create, and why is one bin omitted?
- 2 Write the fitted model with an intercept.
- 3 With  $\hat{\beta}_0 = 40$ ,  $\hat{\beta}_1 = 15$  (bin  $[35, 50)$ ),  $\hat{\beta}_2 = 25$  (bin  $[50, \infty)$ ), predict wage for ages 30, 40, and 60.
- 4 State one limitation of this approach.

### Live demo — §2 of the notebook

Switch to `chapter_07_lab.ipynb` §2 and run the binning cell: four age bins give mean wages  $94.2 \rightarrow 118.2 \rightarrow 117.8 \rightarrow 101.8$  — the hump no straight line can produce, at the price of a jump at age 33.5.

## Solution — Exercise 7.2 (1/2)

### Solution

**(a) Two dummies for three bins.** Define  $C_1(x) = 1[35 \leq x < 50]$  and  $C_2(x) = 1[x \geq 50]$ . The first bin  $(-\infty, 35)$  is the *reference category*, absorbed into the intercept. *Why omit one bin?* With an intercept, the column of ones equals the sum of all three bin dummies, so including all three would make the design matrix perfectly collinear — the “dummy-variable trap”. In general:  $m$  bins  $\Rightarrow m - 1$  dummies plus an intercept.

**(b) Fitted model.**

$$y = \beta_0 + \beta_1 C_1(x) + \beta_2 C_2(x) + \varepsilon.$$

*Reading the coefficients:*  $\beta_0$  is the mean wage in the reference bin;  $\beta_1$  and  $\beta_2$  are the mean *differences* of bins 2 and 3 relative to that reference — not relative to each other.

## Solution — Exercise 7.2 (2/2)

### Solution

(c) **Predictions.** Locate each age's bin, switch on its dummy, and add:

- Age  $30 < 35$ : reference bin,  $C_1 = C_2 = 0 \Rightarrow \hat{y} = 40$ .
- Age  $40 \in [35, 50)$ :  $C_1 = 1, C_2 = 0 \Rightarrow \hat{y} = 40 + 15 = 55$ .
- Age  $60 \geq 50$ :  $C_1 = 0, C_2 = 1 \Rightarrow \hat{y} = 40 + 25 = 65$ .

*Interpretation:* predicted wage steps from 40 to 55 at age 35 and to 65 at age 50 — everyone inside a bin gets the same prediction.

(d) **Limitation.** The fit is constant within each bin and jumps at the cutpoints: it ignores any trend inside a bin, and the cutpoints are chosen arbitrarily. Unless the true relationship really has breakpoints, this misses smooth structure.

### Common mistake

Reading  $\beta_2 = 25$  as the jump from bin 2 to bin 3. Both coefficients compare to the *reference* bin; the bin-2-to-bin-3 step is  $\beta_2 - \beta_1 = 10$ .

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# Basis functions: a unifying view

Polynomials and step functions are special cases of:

## Basis-function regression

$$Y = \beta_0 + \sum_{j=1}^K \beta_j b_j(X) + \varepsilon,$$

for fixed basis functions  $b_1, \dots, b_K$ .

### How to read this

Read it as: each  $b_j(\cdot)$  is a *fixed* transformation chosen in advance, and  $\beta_j$  is its ordinary regression weight — so once the  $b_j$  are set, this is plain OLS. Special cases:

- Polynomials:  $b_j(X) = X^j$ .
- Step functions:  $b_j(X) = 1\{c_j \leq X < c_{j+1}\}$ .
- Splines: piecewise polynomials joined smoothly at chosen knots.



# Regression splines: idea

Splines combine the flexibility of polynomials with the locality of step functions.

## Construction

Split the range of  $X$  at  $K$  *knots*  $\xi_1 < \dots < \xi_K$ . Fit a polynomial of degree  $d$  on each piece, joined at the knots so the function and its first  $d - 1$  derivatives are continuous.

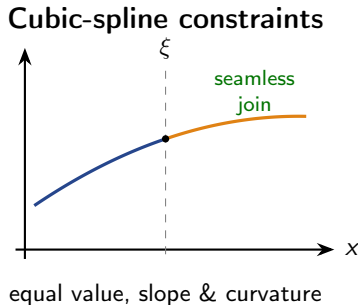
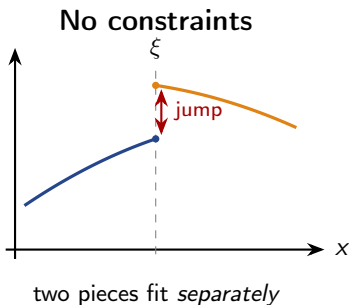
## Practical defaults

**Cubic splines** ( $d = 3$ ) are the standard choice — they are smooth visually. Total degrees of freedom:  $K + d + 1$ .

## Live demo — §3 of the notebook

Switch to `chapter_07_lab.ipynb` §3 and run the `SplineTransformer` cell: a cubic B-spline basis with 6 knots, fitted by plain least squares — the curve bends like the polynomial but stays smooth with no jumps.

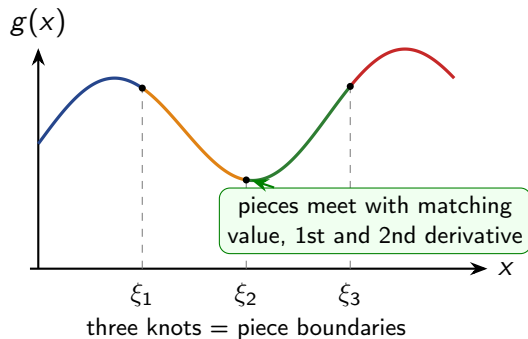
# Why the continuity constraints? Broken vs. smooth joins



## Takeaway

A regression spline is *not* one global polynomial: it is separate low-degree pieces. All the work is in the join — forcing equal value, slope and curvature at each knot makes the seam invisible while each piece stays flexible.

# How a cubic spline is built: continuity at the knots



## Takeaway

Each region is a free cubic, yet at every knot the pieces share value, slope and curvature — so the joins are invisible.

# The natural-spline boundary fix

## Boundary problem

Cubic splines flare beyond extreme knots. The CIs at the boundaries become unreliably wide.

## Why the edges misbehave

Past the outermost knots the fit is still a free cubic, but almost no data pin it down there. A cubic can curve sharply, so a small change in a boundary point swings the whole tail — inflating variance exactly where evidence is thinnest.

## Natural cubic splines

Constrain the spline to be *linear* past the boundary knots. The resulting fits are stable at the edges with negligible bias inside.

# Choosing knots and degrees of freedom

## Two practical decisions

- **How many knots?** Choose by cross-validation. Modest df (4–8) usually suffices.
- **Where?** Equal quantiles of  $X$  usually work well. With small  $n$ , place knots manually at obvious change-points.

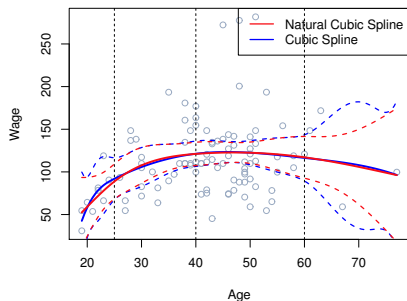
## Why cubic and not quintic?

Cubic is the lowest degree that gives visually smooth fits (matches second derivative). Higher degrees rarely improve the fit but add parameters and instability.

## Takeaway

The knot count is the real tuning parameter: more knots track every wiggle in the data, fewer give a smooth curve that misses some of it. Choose it by cross-validation, not by eye.

# Cubic vs. natural spline on Wage



## Takeaway

Same data, two fits: the cubic spline's error bands blow up at the edges while the natural spline's stay controlled — boundary behaviour is the deciding difference.

## Exercise 7.3 — Degrees of freedom of a cubic spline [Math]

### Exercise

Consider a cubic regression spline with  $K$  interior knots, fitted so that the function and its first two derivatives are continuous at each knot.

- 1 If we fit a *separate* cubic in each of the  $K + 1$  regions, how many free parameters is that?
- 2 How many constraints do the continuity conditions impose in total?
- 3 Show that the net degrees of freedom equal  $K + 4$ .
- 4 How many basis functions does the truncated-power basis use (including the intercept)? Confirm consistency.

### Live demo — §6 of the notebook

Switch to `chapter_07_lab.ipynb` §6 and run the basis-building cell: it constructs the truncated-power basis on Wage ( $K = 3$  knots at the age quartiles) and confirms the design matrix has rank  $7 = K + 4$  — the count you are about to derive.

## Solution — Exercise 7.3 (1/2)

### Solution

**(a) Free parameters.**  $K$  knots split the line into  $K + 1$  regions. Each region carries its own cubic  $a_j + b_jx + c_jx^2 + d_jx^3$ , i.e. 4 coefficients, so fitting them separately costs

$$4 \times (K + 1) = 4K + 4 \text{ parameters.}$$

**(b) Constraints.** At each interior knot we require the two adjacent pieces to agree in *value*, *slope*, and *curvature* ( $g, g', g''$  continuous): 3 constraints per knot, hence

$$3 \times K = 3K \text{ constraints in total.}$$

*Why stop at the second derivative?* Matching  $g'''$  as well would force adjacent cubics to be identical — the “spline” would collapse to one global cubic. Continuity up to  $g''$  is exactly what the eye perceives as smooth.



## Solution — Exercise 7.3 (2/2)

## Solution

(c) **Net degrees of freedom.** Each independent linear constraint removes one free parameter:

$$\underbrace{(4K + 4)}_{\text{free params}} - \underbrace{3K}_{\text{constraints}} = K + 4.$$

*Sanity check:*  $K = 0$  gives 4 — a single cubic, as it must. Each additional knot buys exactly *one* extra degree of freedom.

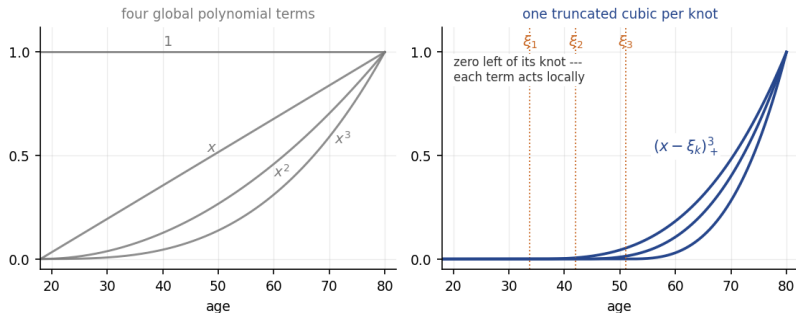
(d) **Basis count.** The truncated-power basis is

$$1, x, x^2, x^3, (x - \xi_1)_+^3, \dots, (x - \xi_K)_+^3,$$

which is  $4 + K$  functions — one regression coefficient each, exactly matching the  $K + 4$  df from (c). Each truncated term adds one knot while preserving continuity of  $g, g', g''$ , so the two counting arguments agree.

*Common mistake:* counting 4 constraints per knot (including  $g'''$ ) or forgetting the intercept among the basis functions — both break the  $K + 4$  tally.

# The truncated-power basis, drawn



## Takeaway

Exercise 7.3's count, drawn: four global polynomial terms plus one truncated cubic per knot ( $K = 3$  knots at the Wage age quartiles)  $= K + 4 = 7$  basis functions. Each truncated term is *zero left of its knot* — that locality is what a polynomial can never have.

## Exercise 7.4 — Natural vs. cubic splines [Concept]

### Exercise

- 1 What extra constraints turn a cubic spline into a *natural* cubic spline?
- 2 A cubic spline with  $K$  knots has  $K + 4$  df. How many df does a natural cubic spline with  $K$  knots have?
- 3 Why do natural splines behave better near the boundaries of the data?
- 4 What does this imply for the width of pointwise confidence bands at the extremes of  $x$ ?

## Solution — Exercise 7.4 (1/2)

### Solution

**(a) Extra constraints.** A natural spline is required to be *linear* beyond the two boundary knots. A function is linear exactly when its second and third derivatives vanish, so the requirement translates into

$$g''(\xi) = 0 \quad \text{and} \quad g'''(\xi) = 0 \quad \text{at and beyond each boundary knot,}$$

i.e. 2 extra constraints at each of the 2 boundaries: 4 additional constraints in total.

**(b) Degrees of freedom.** Start from the cubic-spline count and subtract one df per constraint:

$$(K + 4) - 4 = K.$$

*Interpretation:* a natural spline with  $K$  knots is exactly as complex as a linear model with  $K$  coefficients — its flexibility budget is spent entirely in the interior, where the data live.

## Solution — Exercise 7.4 (2/2)

### Solution

**(c) Why natural splines behave at the boundaries.** Past the outer knots an unrestricted cubic spline is still a free cubic, but almost no data pin it down there: a handful of extreme observations steer four polynomial coefficients, so small data changes swing the whole tail — high variance exactly where evidence is thinnest. Forcing linear tails removes those unsupported cubic wiggles.

**(d) Consequence for confidence bands.** The pointwise band width tracks the variance of  $\hat{g}(x)$ . Because fewer parameters are spent at the extremes and the tail shape is constrained to be linear, natural-spline bands are substantially *narrower* at the boundaries than cubic-spline bands, at the price of a negligible bias increase in the interior.

### Common mistake

Thinking the natural spline is linear *between* the boundary knots too — inside the boundary knots it is every bit as flexible a cubic spline; only the two tails are straightened.

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# Smoothing splines: let the data choose

Instead of choosing knots manually, place a knot at *every* data point and let a penalty control smoothness.

## Penalised RSS

$$\min_g \sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt.$$

## Takeaway

Two terms pull opposite ways: the RSS wants a curve through every point, the penalty wants a straight line.  $\lambda$  sets the exchange rate — and over-smoothing flattens exactly the peaks that matter most.

# Reading the smoothing-spline objective

## How to read this

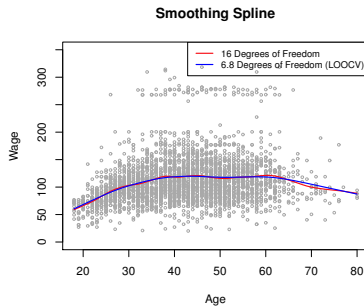
| Term                    | Role                                 |
|-------------------------|--------------------------------------|
| $\sum (y_i - g(x_i))^2$ | Training fit                         |
| $\int g''(t)^2 dt$      | Penalty for “wiggleness” (curvature) |
| $\lambda$               | Smoothness tuning parameter          |

## Three observations

- The minimiser is a natural cubic spline with knots at every  $x_i$ .
- $\lambda = 0$ : interpolation.  $\lambda \rightarrow \infty$ : linear fit.
- “Effective degrees of freedom” parameterise the smoothness.



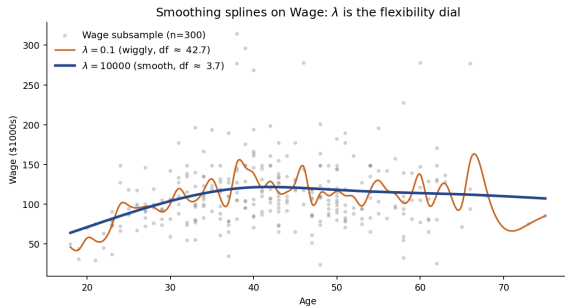
# Smoothing spline on Wage



## Takeaway

Two ways to fix smoothness: a preset 16 df vs. the  $\approx 6.8$  effective df chosen by CV. More effective df means a wigglier fit; CV finds the sweet spot.

# Smoothing spline flexibility: $\lambda$ is the dial



## Takeaway

Same Wage data, two penalties:  $\lambda = 10^4$  gives a calm trend ( $\approx 3.7$  effective df) while  $\lambda = 0.1$  ( $\approx 43$  df) chases noise —  $\lambda$  is the flexibility dial, tuned by CV/LOOCV.

## Local regression (LOESS): a different philosophy

Instead of one global function, fit a small model around each point.

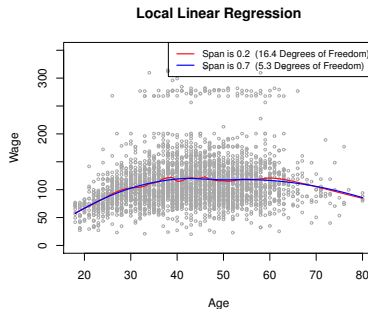
### Recipe at each query point $x_0$

- 1 Find the fraction  $s$  (“span”) of points closest to  $x_0$ .
- 2 Weight them by a kernel that decays with distance.
- 3 Fit a low-degree polynomial by weighted least squares.
- 4 Evaluate at  $x_0$ ; repeat over the grid of  $x_0$ .

### How to read this

Span  $s$  controls smoothness — the analogue of  $\lambda$ . LOESS is excellent for one or two predictors and dies fast as  $p$  grows.

# LOESS on Wage



## Takeaway

Each curve uses a different span: 0.2 tracks local wiggles, 0.7 is smoother. Span plays the role of  $\lambda$ , trading bias against variance.

Source: Figure 7.10 — ISLP, James et al. (2023).

## Mini-check: spline choices

### Quick self-assessment

- 1 Why prefer a natural cubic spline over a regular cubic spline at the boundaries?
- 2 How is a smoothing spline's  $\lambda$  related to its effective degrees of freedom?
- 3 In LOESS, what happens to bias as the span shrinks?

### Answers

1. Natural splines are constrained linear past the boundary knots, avoiding wild boundary oscillation. 2. Larger  $\lambda \Rightarrow$  smaller df (more shrinkage, smoother fit). 3. Bias decreases (the local fit is more local), but variance increases.

## Exercise 7.5 — Smoothing splines and $\lambda$ [Concept]

### Exercise

A smoothing spline chooses  $g$  to minimise

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt.$$

- 1 Interpret the penalty term. What does it measure?
- 2 Describe the fit as  $\lambda \rightarrow 0$  and as  $\lambda \rightarrow \infty$ .
- 3 Define the effective degrees of freedom  $\text{df}_\lambda$  and give its range.
- 4 How is  $\lambda$  chosen in practice?

## Solution — Exercise 7.5 (1/2)

### Solution

**(a) The penalty measures curvature.**  $g''(t)$  is the rate of change of the slope at  $t$ ; squaring removes the sign and integrating adds curvature up over the whole range, so  $\int g''(t)^2 dt$  is the total *roughness* of  $g$ . A straight line has  $g'' \equiv 0$  and pays no penalty; a wiggly curve pays a lot. The penalty therefore discourages wiggleness.

**(b) The two extremes.** As  $\lambda \rightarrow 0$  curvature becomes free, so the minimiser drives the RSS term to 0:  $g$  *interpolates* the data — extremely wiggly, low bias, high variance. As  $\lambda \rightarrow \infty$  any curvature is infinitely expensive, forcing  $g'' = 0$  everywhere, so  $g$  collapses to the ordinary least-squares *line*.

*Interpretation:*  $\lambda$  sweeps a continuous bridge between interpolation and linear regression — it is the bias-variance dial of the smoothing spline.

## Solution — Exercise 7.5 (2/2)

### Solution

**(c) Effective degrees of freedom.** For fixed  $\lambda$  the fit is linear in  $y$ :  $\hat{g} = S_\lambda y$  for a smoother matrix  $S_\lambda$ . In OLS the analogous “hat” matrix has trace equal to the number of parameters, which motivates the definition

$$\text{df}_\lambda = \text{trace}(S_\lambda),$$

ranging from  $n$  (at  $\lambda = 0$ , interpolation) down to 2 (at  $\lambda \rightarrow \infty$ , slope + intercept of the linear fit).

**(d) Choosing  $\lambda$ .** By cross-validation. LOOCV is essentially free because each leave-one-out residual follows from the *single* full fit via the leverage  $\{S_\lambda\}_{ii}$ :

$$\text{RSS}_{\text{cv}}(\lambda) = \sum_{i=1}^n \left( \frac{y_i - \hat{g}_\lambda(x_i)}{1 - \{S_\lambda\}_{ii}} \right)^2.$$

### Common mistake

Picking  $\lambda$  by minimising *training* RSS — that always selects  $\lambda = 0$  (interpolation). Only out-of-sample criteria can trade fit against roughness.



# Outline

- 1 Why this chapter matters
- 2 Polynomials
- 3 Splines
- 4 Smoothing
- 5 GAMs**
- 6 Python Lab
- 7 Summary

# From one predictor to many: GAMs

Everything we have done so far is for one predictor at a time. Generalised additive models lift the techniques into the multivariate setting.

## The GAM model

$$Y = \beta_0 + \sum_{j=1}^p f_j(X_j) + \varepsilon,$$

where each  $f_j$  is a smooth function (spline or LOESS).

## Symbol by symbol

$\beta_0$  is the overall level; each  $f_j$  is a smooth, data-driven curve for predictor  $X_j$ ; summing them keeps the model additive. Setting  $f_j(X_j) = \beta_j X_j$  recovers ordinary linear regression — the GAM simply lets each term bend.

# Why GAMs are so useful

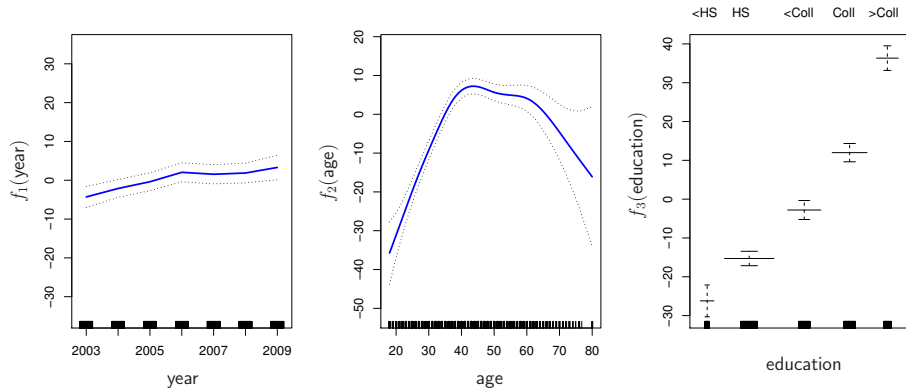
## Three reasons

- ① Generalises linear regression:  $f_j(X_j) = \beta_j X_j$  is the linear special case.
- ② Avoids the curse of dimensionality via the additive structure.
- ③ Each  $f_j$  can be plotted separately — excellent interpretability.

## Limitation

By construction, GAMs miss interactions. Add tensor-product terms  $f_{jk}(X_j, X_k)$  explicitly, or move to trees (Ch. 8).

# GAM fit on Wage

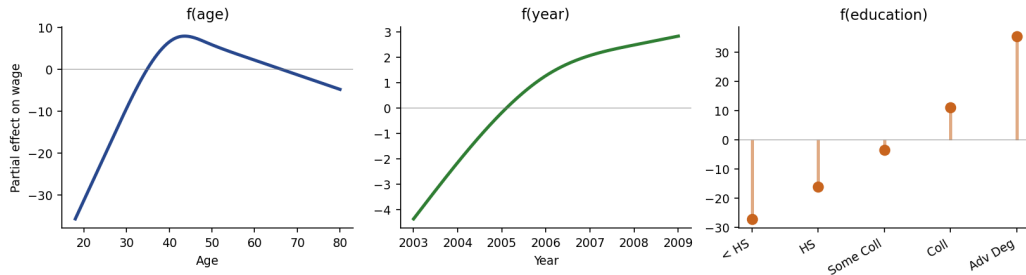


Each panel: estimated  $f_j$  for a different predictor, with pointwise confidence bands.

Source: Figure 7.12 — ISLP, James et al. (2023).

# GAM component functions fitted on Wage

GAM component functions:  $\text{wage} \sim s(\text{age}) + s(\text{year}) + \text{education}$



## Takeaway

Reading each partial effect: wage rises with age then falls after  $\sim 60$ ; it drifts up gently with year; and it increases monotonically with education. The additive structure lets us plot and interpret one predictor at a time.

# GAMs for classification

Same idea inside a GLM:

## Logistic GAM

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + \sum_{j=1}^p f_j(X_j).$$

## Reading the model

The left side is the *log-odds* of the positive class; each smooth  $f_j$  shifts those log-odds up or down as  $X_j$  changes. It is an additive logistic model — interpret every  $f_j$  on its own, just as in a linear GLM.

Combines logistic regression's interpretability with smooth per-predictor effects.

Implementations: `pyGAM`, `statsmodels.gam`.

## Extended Exercise 7.2 — A GAM for wage [Integrative]

### Extended exercise (15 min)

Fit a generalised additive model

$$\text{wage} \approx \beta_0 + s(\text{age}) + \text{year} + f(\text{education}),$$

with a smooth spline in age, a (near-)linear term in year, and education as a factor.

- 1 Fit it (e.g. pyGAM) and state how a GAM is estimated.
- 2 Interpret each partial effect.
- 3 Which term is genuinely non-linear? Discuss the effective degrees of freedom (EDF) of each term.
- 4 State the advantages of the additive form and its key limitation. Give an interaction it would miss and how to add it.

## Solution — Extended Exercise 7.2 (1/3)

## Solution

**(a) Fit & algorithm.** Encode education to integer codes, then fit a spline in age, a linear term in year and a factor in education:

```
import pandas as pd
from pygam import LinearGAM, s, l, f

W = pd.read_csv('../ALL CSV FILES - 2nd Edition/Wage.csv') # load Wage
W['edu'] = W['education'].astype('category').cat.codes # levels -> 0..4
X = W[['age', 'year', 'edu']].values # cols 0, 1, 2 in that order
y = W['wage'].values # response
gam = LinearGAM(s(0) + l(1) + f(2)).fit(X, y) # smooth, linear, factor
gam.summary() # EDF per term: age ~15, year ~1, edu 4
```

A GAM is fit by *backfitting*: cycle through the terms, repeatedly re-estimating each  $f_j$  on the partial residual  $y - \beta_0 - \sum_{k \neq j} f_k(X_k)$  until the fit stabilises — each smooth is refitted to whatever the other terms leave unexplained.



## Solution — Extended Exercise 7.2 (2/3)

### Solution

(b) **Partial effects** (each holds the others fixed):

- age: a pronounced *hump* — wage climbs steeply to about age 40, plateaus through the 50s, then declines after  $\sim 60$ .
- year: a gentle, near-linear *increase* — slow wage growth over the sample window.
- education: a monotone *staircase* — each higher band pays more, “Advanced Degree” far above “<HS Grad”.

(c) **Non-linearity & EDF.** Only age is genuinely non-linear: its smooth spends many effective degrees of freedom ( $\approx 15$  at pyGAM’s default penalty, dropping to  $\approx 10$  once  $\lambda$  is tuned by `gam.gridsearch`). year enters with EDF  $\approx 1$  (essentially a straight line); education as a factor contributes  $df = (\text{levels} - 1) = 4$ . The EDF is how a GAM quantifies “how wiggly” each term is.

### Common mistake

Reading a partial-effect curve as an absolute wage level. Each panel is centred around zero; only its *shape* and within-panel differences are meaningful, with all other predictors held fixed.

## Solution — Extended Exercise 7.2 (3/3)

### Solution

#### (d) Advantages of additivity.

- Each term is estimated and *plotted separately*, so effects are directly interpretable.
- The additive structure sidesteps the curse of dimensionality (no  $p$ -dimensional smoother).
- It strictly generalises linear regression: a linear term is the special case  $f_j(X_j) = \beta_j X_j$ .

**Key limitation — no interactions.** By construction the shape of  $s(\text{age})$  is the *same* at every education level. If the wage–age profile were steeper for college graduates than for high-school graduates (an  $\text{age} \times \text{education}$  interaction), the additive GAM could not represent it. *Fix:* add a bivariate/tensor term  $f(\text{age}, \text{education})$ , fit separate age-smooths per group, or move to trees / boosting (Ch. 8), which capture interactions automatically.

# Outline

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# Python lab — run it live in the notebook

## Companion notebook: `chapter_07_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell, section by section:

- §1, *Polynomial regression*, and §2, *Step functions*;
- §3, *Regression splines*, and §4, *LOESS*;
- §5, *GAM*, and §6, *Where the degrees of freedom come from*.

Data loads via the ISLP package or the bundled CSV files; §7, *Exercises*, closes the notebook.

# Polynomial regression

```
import statsmodels.api as sm
from ISLP import load_data
from ISLP.models import poly, ModelSpec as MS

Wage = load_data("Wage") # 3000 workers: wage, age, year
design = MS([poly("age", degree=4)]).fit_transform(Wage) # 4 age cols
res = sm.OLS(Wage["wage"], design).fit() # plain OLS on that design
print(res.summary()) # one coefficient per basis column
```

## What the design matrix actually holds

`fit_transform` is what builds the columns OLS sees: an intercept plus *four* columns for age. `poly` returns *orthogonal* polynomials — rotated combinations of  $\text{age}$ ,  $\text{age}^2$ ,  $\text{age}^3$ ,  $\text{age}^4$  that span the same space, so the fitted curve and  $R^2 \approx 0.086$  match raw powers exactly, but the columns are uncorrelated. Judge the term by the fitted curve, never by one column's  $t$ -statistic.

# Regression splines

```
from ISLP.models import bs, ns

design = MS([bs("age", df=6)]).fit_transform(Wage) # 6 cubic cols
design2 = MS([ns("age", df=4)]).fit_transform(Wage) # 4 cols, linear tails
res = sm.OLS(Wage["wage"], design2).fit() # still ordinary least squares
print(res.summary()) # one weight per basis function
```

## df and knots are two ways to say the same thing

Each call returns a matrix whose columns are the basis functions  $b_j(\text{age})$  of the slide before. `df` says *how many columns*; the knots are then placed for you at quantiles of age. Naming the knots instead gives the identical fit, because cubic-spline  $\text{df} = K + 4$  and natural-spline  $\text{df} = K$  — which is why `ns` at the same `df`, spending nothing on the tails, can afford more knots where the data actually are.

# Local regression with statsmodels

```
import statsmodels.nonparametric.api as nps

l = nps.lowess(endog=Wage["wage"], exog=Wage["age"], # y first, then x
              frac=0.2, return_sorted=True) # span: 20% of the points

import matplotlib.pyplot as plt
plt.scatter(Wage["age"], Wage["wage"], s=4, alpha=0.3) # raw data
plt.plot(l[:, 0], l[:, 1], color="orange") # col 0 = age, col 1 = fit
plt.show()
```

No coefficients come back — just the curve

One  $(x, \hat{y})$  row per point, sorted by  $x$  so it can be plotted.

Live demo — §4 of the notebook

Switch to chapter\_07\_lab.ipynb §4 and run the lowess cell: with span 0.2 each local fit uses 20% of the 3000 workers, and the curve traces the same rise–plateau–fall as the splines.

# A simple GAM with pyGAM

```
from pygam import LinearGAM, s
```

```
X = Wage[["year", "age"]].values # col 0 = year, col 1 = age
```

```
y = Wage["wage"].values # response
```

```
gam = LinearGAM(s(0) + s(1)).fit(X, y) # s(j): smooth of column j
```

```
gam.summary() # EDF per term: year ~7, age ~14
```

## Reading the EDF column

`s(0)` and `s(1)` index *columns* of `X`. A term's effective df says how much flexibility the penalty let it use: 1 is a straight line, year  $\approx 7$  a gentle drift, age  $\approx 14$  the pronounced hump.

## Live demo — §5 of the notebook

Switch to `chapter_07_lab.ipynb` §5 and run the two GAM cells: the summary reports  $\approx 21.2$  total effective df (age's smooth alone spends  $\approx 14$ ), and the partial-dependence panels show the age hump next to the gentle year drift.



## Exercise 7.6 — Natural spline / GAM on Wage [Python]

### Exercise

Using `Wage.csv`:

- 1 Fit wage on a natural cubic spline of age with 5 df via OLS.
- 2 Fit a GAM  $\text{wage} \sim s(\text{age}) + s(\text{year})$ .
- 3 Describe the estimated effect of age.

### Run this live

The worked solution sits in `chapter_07_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Exercise 7.6 — Natural spline / GAM on Wage [Python]*.

## Solution — Exercise 7.6 (1/3)

## Solution

(a) Natural cubic spline via patsy's `cr()` basis — `cr()` expands age into 5 natural-spline columns (linear tails built in), so the fit is still ordinary OLS:

```
import pandas as pd
import statsmodels.formula.api as smf

Wage = pd.read_csv('../ALL CSV FILES - 2nd Edition/Wage.csv') # load
fit = smf.ols('wage ~cr(age, df=5)', data=Wage).fit() # 5 ns columns
print(fit.summary()) # read the joint F, not single t's
```

What the output shows: 6 coefficients (intercept + 5 basis columns),  $R^2 \approx 0.087$ , and an overall  $F$ -test with  $p \approx 6 \times 10^{-58}$  — the spline terms are jointly highly significant, even though age alone explains only a modest share of wage variance. `df=5` is the *entire* complexity choice: it fixes how many basis columns are built, and the knots then go at quantiles of age.

## Solution — Exercise 7.6 (2/3)

### Solution

(b) A GAM with smooth terms via pyGAM — each `s()` is a penalised spline whose wiggleness is measured in effective degrees of freedom (EDF):

```
import numpy as np
from pygam import LinearGAM, s

X = Wage[['age', 'year']].values # col 0 = age, col 1 = year
y = Wage['wage'].values # response
gam = LinearGAM(s(0) + s(1)).fit(X, y) # a penalised spline per column
gam.summary() # EDF per term: how wiggly each one got
```

What the output shows: with pyGAM's default penalty the age smooth uses  $\approx 15$  EDF and year  $\approx 6$  — both flexible; tuning  $\lambda$  by `gam.gridsearch(X, y)` shrinks the smooths toward simpler shapes. Note the contrast with (a): there you fixed  $df = 5$ ; here  $\lambda$  decides, and the EDF merely reports the flexibility it granted.

## Solution — Exercise 7.6 (3/3)

### Solution

(c) *Interpretation.* The fitted effect of age is clearly nonlinear: wage rises steeply from the late teens to about age 40, is roughly flat (a plateau) between 40 and 60, then declines gently after 60. A straight line would miss this rise-then-plateau-then-fall shape, which is why a spline / GAM is preferable. The year effect is mildly increasing (slow wage growth over time).

*How to read the fit:* plot `gam.partial_dependence()` per term rather than staring at coefficients — the spline basis coefficients have no individual meaning; only the fitted curve does.

### Common mistake

Interpreting single spline-basis coefficients (or their *t*-tests) as “the effect of age”. Judge the term by its fitted curve and a *joint* test of all its basis columns.

## Extended Exercise 7.3 — Three fits for wage vs. age [Python]

### Extended exercise (15 min)

Using `Wage.csv`, model wage as a function of age three ways and compare:

- 1 a **polynomial**, choosing the degree by nested ANOVA  $F$ -tests (or 5-fold CV);
- 2 a **step function** (bin age into several ranges);
- 3 a **natural cubic spline** with about 5 df.

Overlay the three fitted curves on a scatterplot, report the chosen complexity, and interpret where the fits agree and where they differ.

### Run this live

The worked solution sits in `chapter_07_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 7.3 — Three fits for wage vs. age [Python]* — the nested ANOVA, the three fits and the overlay plot, ready to run.

## Solution — Extended Exercise 7.3 (1/3)

## Solution

Choose the polynomial degree by nested ANOVA.

```
import numpy as np, pandas as pd
import statsmodels.formula.api as smf
from statsmodels.stats.anova import anova_lm

W = pd.read_csv('../ALL CSV FILES - 2nd Edition/Wage.csv') # load Wage
terms = lambda d: '+' .join(f'np.power(age,{k})' for k in range(1, d+1)) # age..age^d
models = [smf.ols('wage ~'+terms(d), data=W).fit() for d in range(1, 6)] # nested: +1 term each
print(anova_lm(*models)[['df_diff', 'F', 'Pr(>F)']].round(4)) # row d: is degree d+1 worth it?
```

Each row of `anova_lm` tests degree  $d$  against  $d+1$  — valid because the models are *nested*. The  $F$ -tests are highly significant up to degree 3 ( $2 \rightarrow 3$ :  $p \approx 0.002$ ), *borderline* for  $3 \rightarrow 4$  ( $p \approx 0.05$ ) and clearly insignificant for  $4 \rightarrow 5$  ( $p \approx 0.37$ ). 5-fold CV of test MSE is essentially flat over degrees 3–4. **Chosen degree: 3 (cubic).**

# Solution — Extended Exercise 7.3 (2/3)

## Solution

Step function, natural spline, and the overlay.

```
import matplotlib.pyplot as plt
poly = smf.ols('wage ~'+terms(3), data=W).fit() # the degree chosen above
W['bin'] = pd.cut(W['age'], [17,25,35,45,55,65,81]) # 6 bins -> 5 dummies
step = smf.ols('wage ~C(bin)', data=W).fit() # a constant inside each bin
ns = smf.ols('wage ~cr(age, df=5)', data=W).fit() # 5 cols, linear tails

g = pd.DataFrame({'age': np.arange(18, 81)}) # one row per age to plot
g['bin'] = pd.cut(g['age'], [17,25,35,45,55,65,81]) # SAME cutpoints
plt.scatter(W['age'], W['wage'], s=4, alpha=.15, color='grey') # data
for m, lab in [(poly, 'poly (d=3)'), (step, 'step'), (ns, 'natural spline')]:
    plt.plot(g['age'], m.predict(g), label=lab) # three curves, one grid
plt.legend(); plt.xlabel('age'); plt.ylabel('wage'); plt.show()
```

*Why the grid carries both columns:* predict re-applies each formula, so g must contain age (for poly, ns) and the matching bin labels (for step), built with the same cutpoints as in training.

## Solution — Extended Exercise 7.3 (3/3)

### Solution

#### Reading the overlay.

- *Where they agree:* in the data-dense interior (ages  $\sim 25$ – $70$ ) all three fits trace the same rise–plateau–fall — wage climbs steeply to about age 40, is roughly flat through the 50s, then eases down.
- *Where they differ:* at the sparse boundaries. The polynomial can bend sharply at the extremes; the step function is blocky and *discontinuous* at the cut-points; the natural cubic spline stays smooth and, being linear beyond its outer knots, is the most stable at the edges.

**Chosen complexity:** a cubic polynomial (degree 3–4), a 6-bin step function, and a natural spline with  $\sim 5$  df — all three explain a similar share of variance ( $R^2 \approx 0.08$ – $0.09$ ), but the natural spline is preferred for its boundary stability.

### Common mistake

Picking the degree that maximises training  $R^2$  — it rises mechanically with every added term; only the nested  $F$ -tests or CV can say when the gain is real.



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- 1 Why this chapter matters
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# Chapter 7 in one slide

## What we accomplished

A toolbox for adding flexibility to a regression while keeping the additive, interpretable structure of a linear model.

- Polynomials, step functions, regression splines, smoothing splines, LOESS.
- GAMs as the natural multivariate extension.
- Cross-validation chooses the smoothness in every method.

# Key formulas at a glance

**Polynomial**  $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$ .

**Step function**  $y_i = \beta_0 + \sum_{k=1}^K \beta_k C_k(x_i) + \varepsilon_i$ , with bins  $C_k$ .

**Cubic spline** truncated-power basis  $1, x, x^2, x^3, (x - \xi_k)_+^3$ ;  $K$  knots  $\Rightarrow$   $df = K + 4$ .

**Natural spline** linear beyond the boundary knots;  $df = K$ .

**Smoothing spline**  $\min_g \sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$ .

**Effective df**  $df_\lambda = \text{trace}(S_\lambda)$ , where  $\hat{g} = S_\lambda y$ ; larger  $\lambda \Rightarrow$  *smaller*  $df_\lambda$ : from  $n$  (at  $\lambda = 0$ , interpolation) down to 2 (as  $\lambda \rightarrow \infty$ , a straight line).

**GAM**  $y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i$ .

**Backfitting** re-estimate each  $f_j$  in turn on the partial residual  $y - \beta_0 - \sum_{k \neq j} f_k(X_k)$ , cycling through the terms until convergence.

# Vocabulary checklist

| Term                         | Meaning                                              |
|------------------------------|------------------------------------------------------|
| Basis function               | Fixed transformation $b_j(X)$ entering linearly      |
| Knot                         | Breakpoint where pieces of a spline meet             |
| Cubic / natural spline       | Degree-3 piecewise polynomial / linear at boundaries |
| Smoothing spline             | Penalised second-derivative fit                      |
| Effective degrees of freedom | Trace of the smoother matrix                         |
| LOESS / local regression     | Weighted local polynomial                            |
| Span                         | Fraction of nearest neighbours used in LOESS         |
| GAM                          | $Y = \beta_0 + \sum_j f_j(X_j) + \varepsilon$        |
| Backfitting                  | Iterative algorithm for fitting GAMs                 |

# Methods comparison

| Method           | Strength                 | Weakness                         |
|------------------|--------------------------|----------------------------------|
| Polynomial       | Easy, captures curvature | Boundary oscillation             |
| Step function    | Interpretable            | Discontinuous                    |
| Cubic spline     | Smooth, controllable df  | Boundary blow-up                 |
| Natural spline   | Stable boundaries        | Tiny extra bias inside           |
| Smoothing spline | Auto-fitting smoothness  | Black-box df                     |
| LOESS            | Locally adaptive         | Curse of dimensionality          |
| GAM              | Additive, interpretable  | Misses interactions unless added |

## Decision rules of thumb

- For a single  $X$  and modest curvature: polynomial degree 2–4.
- For a single  $X$  with multiple regimes: regression splines with a handful of knots placed at quantiles.
- For boundary stability: *natural* cubic splines.
- For multiple predictors with little interaction: a GAM.
- Choose smoothness (df,  $\lambda$ , span) by cross-validation.
- For visible interactions: add tensor-product terms or move to trees (Ch. 8).

# Common pitfalls

## Don't

- 1 Use polynomials of degree  $> 4$  on real data.
- 2 Place knots arbitrarily without checking residuals.
- 3 Forget that LOESS fails fast as  $p$  grows.
- 4 Compare GAM df with linear-regression df one-for-one.
- 5 Trust pointwise CIs of smoothing splines for inference.
- 6 Ignore extrapolation — splines outside the data range are unreliable.

## Connections to the rest of the course

- Ch. 3 is the special case where every  $f_j$  is linear.
- Ch. 5 chooses  $df / \lambda / \text{span}$  by CV.
- Ch. 6 ridge / lasso can be applied to spline basis expansions.
- Ch. 8 trees are the nonparametric alternative when interactions dominate.
- Ch. 10 deep nets generalise GAMs by learning the basis.



## Self-check questions

Each question names the slide that answers it — a wrong answer tells you where to read.

- 1 Why is a cubic spline preferred over a quintic spline? (p. 30)
- 2 Argue intuitively why smoothing splines have a unique minimiser. (p. 41)
- 3 How does the smoothing-spline penalty  $\lambda$  relate to df? (p. 43)
- 4 Sketch a GAM with two predictors. Where would you add an interaction? (p. 52)
- 5 For LOESS, what happens to bias and variance as the span shrinks? (p. 46)
- 6 In the Wage data, would you expect a linear or a spline term in age to win? Why? (p. 32)

## Seven things to remember

- 1 Polynomials = linear regression with  $X^j$  basis.
- 2 Step functions = piecewise constant.
- 3 Cubic splines = piecewise cubics joined smoothly.
- 4 Natural splines stabilise the boundaries.
- 5 Smoothing splines: penalised second derivative.
- 6 LOESS: weighted local polynomials.
- 7 GAMs:  $\sum_j f_j(X_j)$  — flexible *and* interpretable.

# What's next

## Chapter 8: *Tree-Based Methods*.

- Decision trees for regression and classification.
- Bagging, random forests, boosting, BART.
- The dominant family of methods for tabular data.

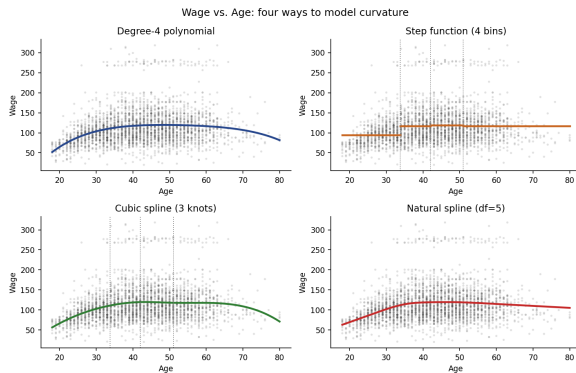
## Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

### Contents of the appendix

- **Wage vs. age: four models side by side** — a second illustration of the cubic-vs-natural contrast already made in the main deck.
- **How the spline basis is built** — the truncated-power term  $(x - \xi)_+^3$  and the constraint counting that turns loose cubic pieces into one smooth curve. The main deck keeps the picture (“Why the continuity constraints?”) and the degrees-of-freedom rule, which is what you are asked to apply.
- **Regression splines by hand (Extended Exercise 7.1)** — evaluating a spline basis and its fit with pencil and paper. Exercise 7.3 covers the same counting argument in five minutes.

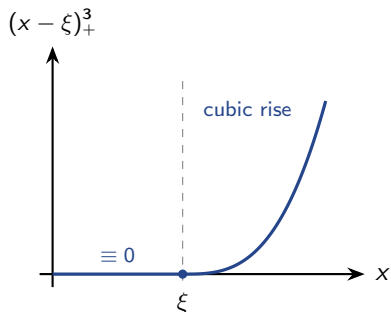
# Wage vs. age: four models side by side



## Takeaway

All four recover the same rise–plateau–fall shape but differ at the edges: the polynomial dips, the step function is blocky, the natural spline stays smooth and stable.

# The building block: one truncated-power term per knot



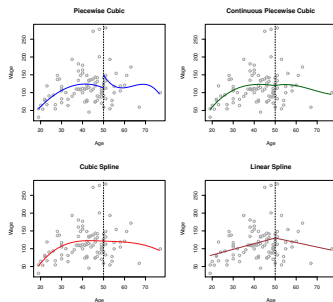
## Reading $(x - \xi)_+^3$

The “+” is the *positive part*: the term is 0 for  $x \leq \xi$  and equals  $(x - \xi)^3$  afterwards. Its value, slope and curvature are all 0 at  $\xi$ , so switching it on adds a bend *only to the right* of the knot without breaking  $C^2$ -smoothness.

## Takeaway

Full basis for  $K$  knots:  $1, x, x^2, x^3, (x - \xi_1)_+^3, \dots, (x - \xi_K)_+^3$  — that is  $K + 4$  functions, exactly the  $K + 4$  degrees of freedom of a cubic spline.

# Adding constraints turns loose pieces into a spline



## Takeaway

Top-left: unconstrained pieces jump. Adding continuity of value, then slope, then curvature (bottom-left) removes the kinks — the cubic spline is smoothest.

Source: Figure 7.3 — ISLP, James et al. (2023).

## Extended Exercise 7.1 — Regression splines by hand [Math]

## Extended exercise (15 min)

Build a cubic regression spline for wage on age with  $K = 3$  interior knots  $\xi_1 = 25 < \xi_2 = 40 < \xi_3 = 60$ .

- 1 Write the truncated-power basis and state the total degrees of freedom.
- 2 The fitted function is

$$\hat{g}(x) = 40 + 0.5x + 0.0004(x - 25)_+^3 - 0.0002(x - 40)_+^3 + 0.0003(x - 60)_+^3.$$

Evaluate  $\hat{g}(50)$ .

- 3 Show that  $\hat{g}, \hat{g}', \hat{g}''$  are continuous at  $\xi_2 = 40$  but  $\hat{g}'''$  jumps.
- 4 How many degrees of freedom does a *natural* cubic spline with the same three knots use? Explain the difference.



## Solution — Extended Exercise 7.1 (1/3)

## Solution

(a) **Truncated-power basis.** With  $K = 3$  knots the cubic-spline basis is

$$1, x, x^2, x^3, (x - \xi_1)_+^3, (x - \xi_2)_+^3, (x - \xi_3)_+^3,$$

so a fitted spline is  $\hat{g}(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \sum_{k=1}^3 \beta_{3+k} (x - \xi_k)_+^3$ .

**Degrees of freedom.** Four base terms plus one per knot:  $4 + K = 7$ . Equivalently, a free cubic in each of the  $K + 1 = 4$  regions has  $4 \cdot 4 = 16$  parameters, minus 3 continuity constraints ( $g, g', g''$ ) at each of the 3 knots:  $16 - 9 = 7$ .

*Why this basis works:* each  $(x - \xi_k)_+^3$  has value, slope and curvature 0 at its knot, so it bends the fit only to the *right* of  $\xi_k$  without breaking smoothness — plain OLS on these 7 columns automatically yields a  $C^2$  piecewise cubic.

# Solution — Extended Exercise 7.1 (2/3)

## Solution

**(b) Evaluate  $\hat{g}(50)$ .** Here the quadratic and cubic *base* coefficients happen to be zero, so only the linear base term and the *active* truncated terms contribute. At  $x = 50$  the knots 25 and 40 are exceeded but 60 is not:

$$(50 - 25)^3 = 25^3 = 15625, \quad (50 - 40)^3 = 10^3 = 1000, \quad (50 - 60)_+^3 = 0.$$

Therefore

$$\begin{aligned}\hat{g}(50) &= 40 + 0.5(50) + 0.0004(15625) - 0.0002(1000) + 0.0003(0) \\ &= 40 + 25 + 6.25 - 0.20 + 0 = 71.05.\end{aligned}$$

Only the basis functions “switched on” at  $x = 50$  enter the sum — this locality is the point of the truncated basis.

## Common mistake

Evaluating the third term as  $(50 - 60)^3 = (-10)^3 = -1000$  instead of 0. The “+” truncation zeroes the term for  $x \leq 60$ ; forgetting it would give the wrong value  $71.05 - 0.30 = 70.75$ .

## Solution — Extended Exercise 7.1 (3/3)

## Solution

(c) **Continuity at  $\xi_2 = 40$ .** Only the term  $t(x) = (x - 40)_+^3$  changes behaviour there. With  $t' = 3(x - 40)_+^2$  and  $t'' = 6(x - 40)_+$ , its one-sided limits at  $x = 40$  are

|                      | $t$ | $t'$ | $t''$ |
|----------------------|-----|------|-------|
| $x \rightarrow 40^-$ | 0   | 0    | 0     |
| $x \rightarrow 40^+$ | 0   | 0    | 0     |

so  $\hat{g}, \hat{g}', \hat{g}''$  are continuous. But  $t''' = 6 \cdot 1\{x > 40\}$  jumps from 0 to 6, hence  $\hat{g}'''$  is discontinuous — the signature of a cubic spline (continuous curvature, kinked third derivative).

(d) **Natural cubic spline.** It is forced *linear* beyond the two boundary knots, imposing 2 extra constraints at each end ( $g'' = g''' = 0$ ), i.e. 4 fewer df:  $7 - 4 = 3$ . A natural spline spends its budget in the interior and is far more stable at the edges.